CIFAR-10 K-Nearest Neighbors Results

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The best setting chosen by cross-validation is **L1 distance with k=7**. Its test accuracy is **39.2%**, 3.9 times random guessing (10%).

Accuracy	Precision (macro)	Recall (macro)	F1-score (macro)
39.2%	**46.3%**	**39.2%**	**38.7%**

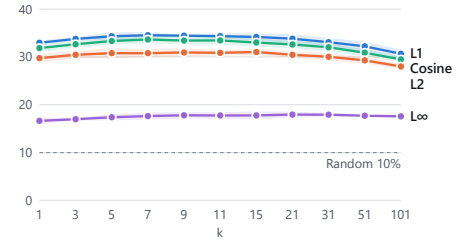
Assignment requirements

- ✓ **KNN implementation** · written in NumPy (knn_cifar10.py)
- ✓ **Values of k** · 11 values: 1, 3, 5, 7, 9, 11, 15, 21, 31, 51, 101
- ✓ **Distance metrics** · L1, L2, L∞, cosine
- ✓ **5-fold cross-validation** · 44 combinations, stratified folds
- ✓ **Evaluation metrics** · Accuracy, Precision, Recall, F1
- ✓ **Comparison and explanation** · see Interpretation below

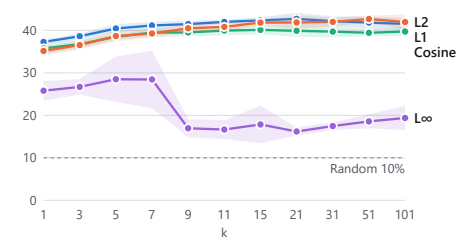
5-fold cross-validation mean by k and distance metric (%)

L1 Cosine L2 L∞ Shading: ±1 standard deviation across folds Circle on F1: chosen k

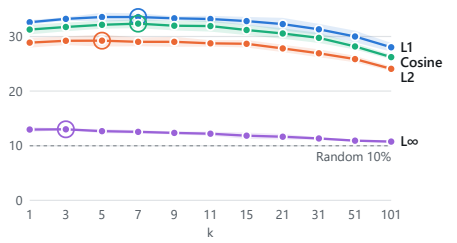
Accuracy = Recall Best: L1, k=7, 34.5%



Precision Best: L1, k=21, 42.7%



F1-score Best: L1, k=7, 33.6%



Best k and performance by distance metric (%)

Distance	Chosen k	5-fold cross-validation (20,000 images)				Test (50,000 train / 10,000 test)			
		Acc	Prec	Rec	F1	Acc	Prec	Rec	F1
L1	**7**	**34.5** ±0.6	**41.2** ±0.9	**34.5** ±0.6	**33.6** ±0.7	**39.2**	**46.3**	**39.2**	**38.7**
Cosine	7	33.7 ±0.8	39.5 ±1.0	33.7 ±0.8	32.4 ±0.8	38.1	44.0	38.1	37.2
L2	5	30.8 ±1.1	38.7 ±1.3	30.8 ±1.1	29.2 ±1.1	35.6	43.7	35.6	34.5
L∞	3	17.0 ±0.3	26.7 ±1.8	17.0 ±0.3	13.0 ±0.1	18.3	27.7	18.3	14.2

k was chosen only by mean cross-validation F1; test results were not used. Because every class has the same number of images, macro recall equals accuracy.

Interpretation

- **Distance metric:** Test accuracy ranks L1 39.2% > cosine 38.1% > L2 35.6% > L∞ 18.3%. L∞ looks only at the single largest pixel difference, discarding the information in the other 3,071 values. L2 squares differences, so a few large pixel differences dominate it, while L1 adds all differences with equal weight. Cosine normalizes vector length, which reduces differences in overall image brightness scale.
- **Size of the differences:** L1 beats L2 by 3.0-4.3 percentage points in all five folds, a clear difference. L1 and cosine differ by only 0.88 points in cross-validation, about the size of the standard deviation across folds, so neither can be called better.
- **Effect of k:** F1 for L1, cosine and L2 peaks at k=5-7 and falls as k grows to 101. k=1 depends on a single nearest image, while a very large k lets unrelated images into the vote.
- **Precision and F1:** As k grows, F1 falls while precision stays high. Predictions concentrate on deer, bird and ship, so the other classes are predicted rarely, which raises their precision and lowers their recall. This is why k was chosen by F1.
- **Differences between classes:** Ship (F1 52.6%) and airplane (50.0%) score highest and cat (26.9%) lowest. Trucks, automobiles and airplanes are often mistaken for ships (244, 219 and 219 images), and frogs and horses for deer (293 and 261 images). This suggests that pixel comparison is drawn to backgrounds and color layout more than to object shape.
- **Overall level:** 39.2% is about four times random guessing, but still low. Raw pixel distances struggle to handle changes in object position, color and background.

Per-class test results for the best setting (L1, k=7)

Per-class scores (%), by F1

Class	Prec	Rec	F1
Ship	41.8	71.2	52.6
Airplane	45.7	55.2	50.0
Horse	65.3	29.4	40.6
Truck	70.4	27.4	39.5
Dog	45.1	31.0	36.7
Frog	37.3	35.1	36.1
Deer	26.7	54.0	35.7
Automobile	69.4	23.8	35.4
Bird	26.7	43.1	33.0
Cat	34.4	22.1	26.9

Confusion matrix (1,000 images per class; darker means more)

True \ Predicted	Airplane	Automobile	Bird	Cat	Deer	Dog	Frog	Horse	Ship	Truck
Airplane	552	6	98	17	47	6	34	6	219	15
Automobile	78	238	85	43	148	52	65	25	219	47
Bird	105	2	431	50	232	31	76	15	55	3
Cat	59	8	163	221	189	127	139	24	61	9
Deer	66	3	223	31	540	20	51	15	51	0
Dog	47	2	177	120	159	310	98	21	59	7
Frog	16	2	212	54	293	32	351	3	35	2
Horse	73	6	135	40	261	61	63	294	50	17
Ship	109	10	34	25	50	20	12	13	712	15
Truck	102	66	54	42	102	29	53	34	244	274

Full cross-validation results (5-fold mean, %)

k	L1				Cosine				L2				L∞			
	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1
1	33.0	37.3	33.0	32.6	31.8	35.8	31.8	31.3	29.7	35.2	29.7	28.9	16.6	25.8	16.6	13.0
3	33.8	38.6	33.8	33.2	32.7	36.8	32.7	31.7	30.5	36.5	30.5	29.2	**17.0**	**26.7**	**17.0**	**13.0**
5	34.3	40.4	34.3	33.6	33.3	38.5	33.3	32.1	**30.8**	**38.7**	**30.8**	**29.2**	17.4	28.5	17.4	12.7
7	**34.5**	**41.2**	**34.5**	**33.6**	**33.7**	**39.5**	**33.7**	**32.4**	30.8	39.3	30.8	29.0	17.6	28.4	17.6	12.5
9	34.4	41.5	34.4	33.3	33.4	39.5	33.4	32.0	30.9	40.5	30.9	29.0	17.8	17.0	17.8	12.3
11	34.4	42.0	34.4	33.2	33.5	40.0	33.5	31.9	30.9	40.8	30.9	28.7	17.7	16.7	17.7	12.2
15	34.2	42.4	34.2	32.8	33.0	40.1	33.0	31.2	31.0	41.8	31.0	28.7	17.8	17.9	17.8	11.8
21	33.8	42.7	33.8	32.3	32.6	39.9	32.6	30.5	30.5	41.9	30.4	27.8	17.9	16.2	17.9	11.6
31	33.1	42.2	33.1	31.3	32.0	39.7	32.0	29.7	30.0	42.0	30.0	26.9	17.9	17.5	17.9	11.3
51	32.2	41.8	32.2	30.0	30.9	39.4	30.9	28.2	29.3	42.7	29.3	25.9	17.7	18.6	17.7	10.9
101	30.7	41.5	30.7	28.0	29.5	39.7	29.5	26.2	28.0	42.0	28.0	24.1	17.6	19.4	17.6	10.8

Blue cells mark the chosen k for each distance metric. Recall equals accuracy because every class has the same number of images. Per-fold values and standard deviations are in results/cv_per_fold.csv and results/cv_summary.csv.

Conditions and limitations

- **Data:** 50,000 training images and 10,000 test images, as 3,072-dimensional raw pixels with no normalization. To limit CPU time, cross-validation used 20,000 images, 2,000 per class (full run: `python knn_cifar10.py --n-cv 50000`).
- **Choosing k:** k was chosen on 20,000 images, so it may differ from the best value for 50,000. For L1, test F1 at k=3 (39.1%) was slightly higher than at the chosen k=7 (38.7%), but test results were not used to change k.
- **L∞:** Its distances are integers with many ties, so results can depend on how tied neighbors are picked.
- **Vote ties:** When votes tie, the class of the nearest neighbor is chosen.
- **Implementation check:** Distances, evaluation metrics and the confusion matrix matched scikit-learn on a sample of real images.